

Lecture-5: Applications DC Machine

Energy Conversion

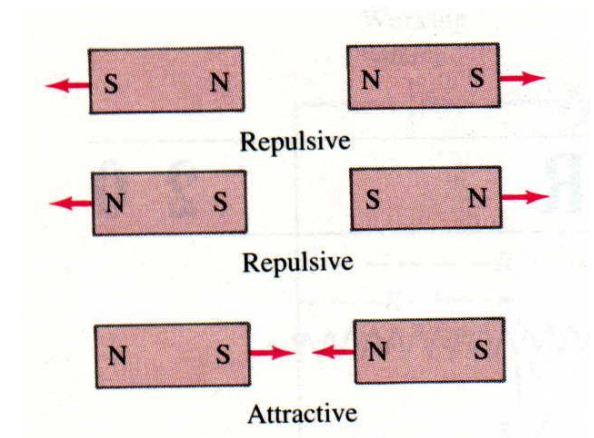
Reza Saeed Kandezy

Outlines

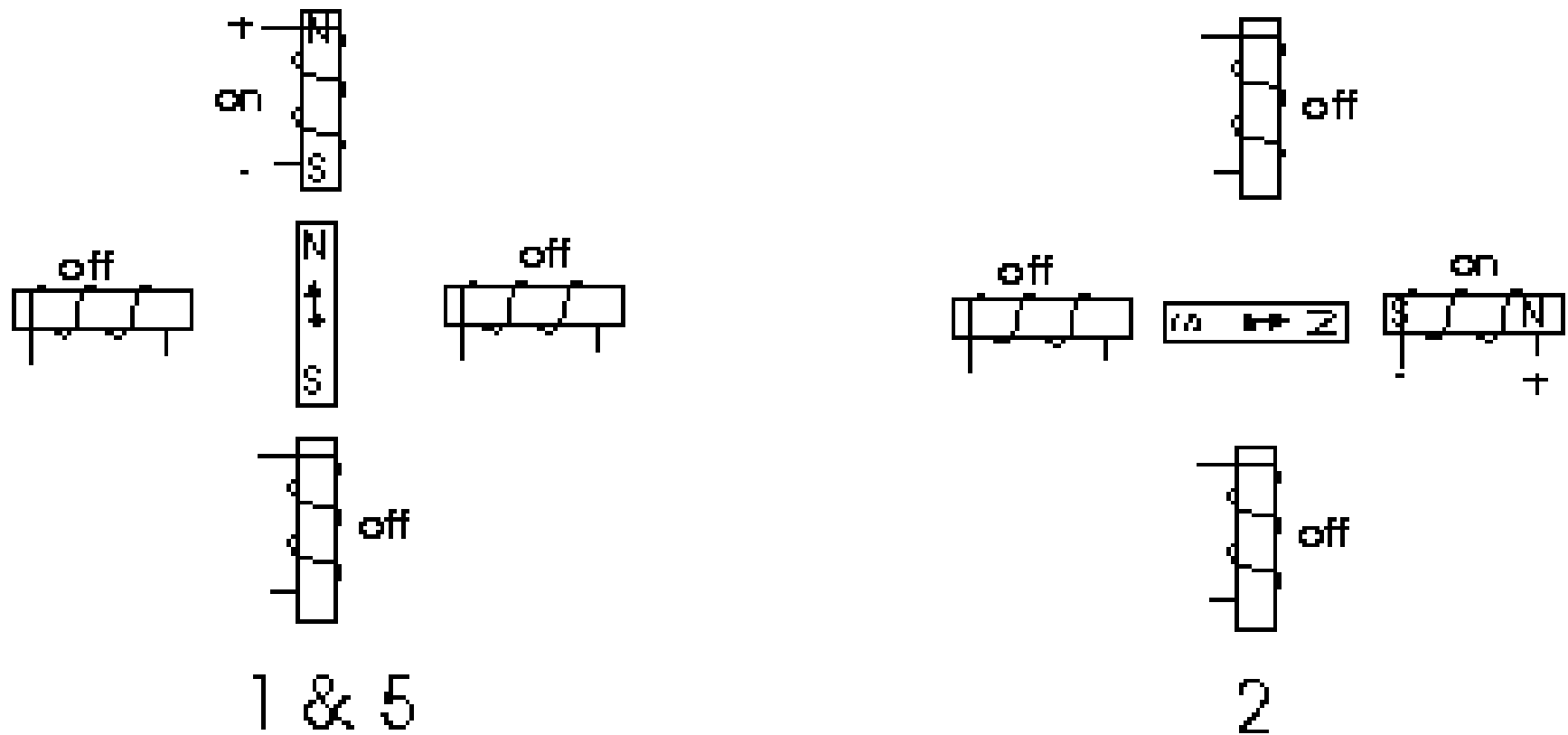
- **Foundation of DC machine**
- **How they work**
- **Equivalent circuits**
- **Analysis of DC machine**

Magnetism

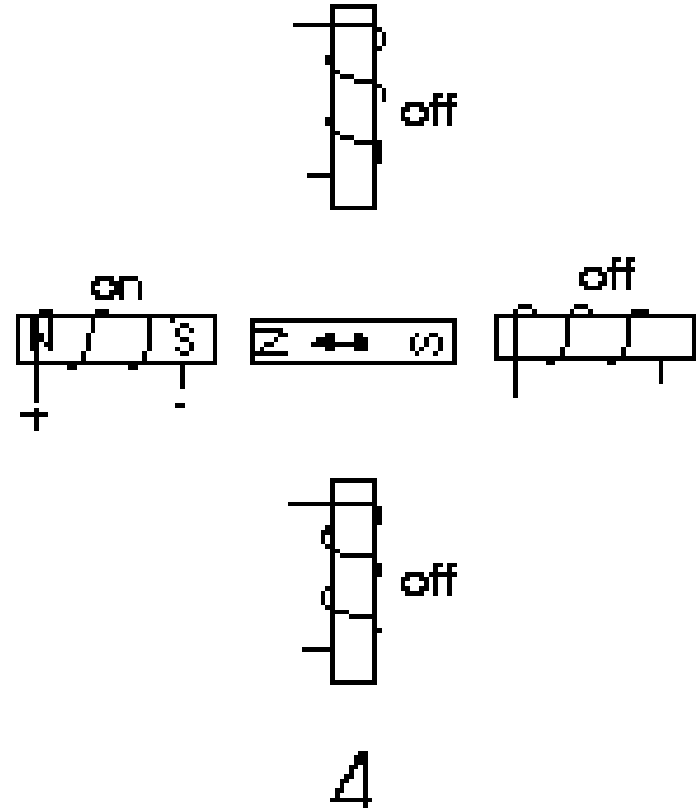
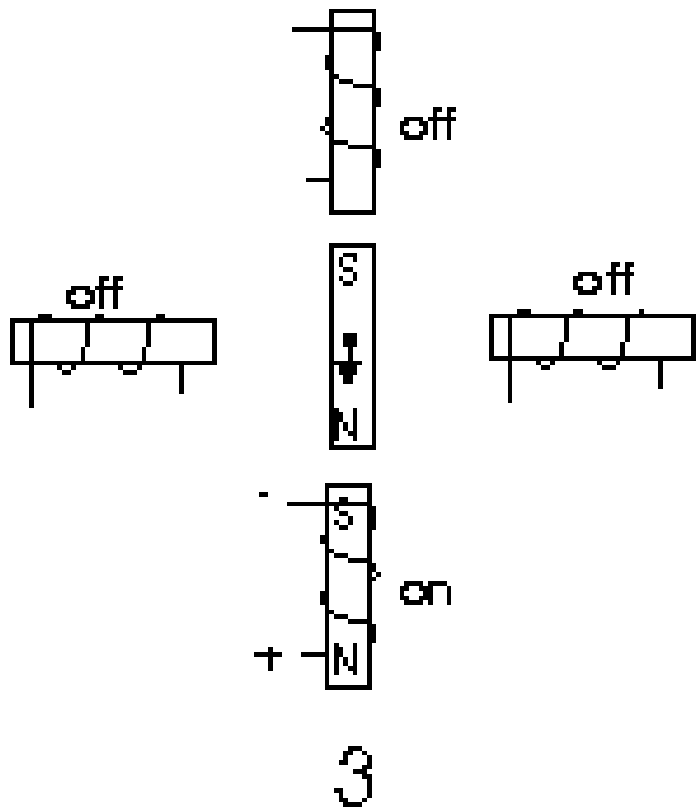
- **Opposites Attract / Likes Repel**



Concept of Stepper Motor

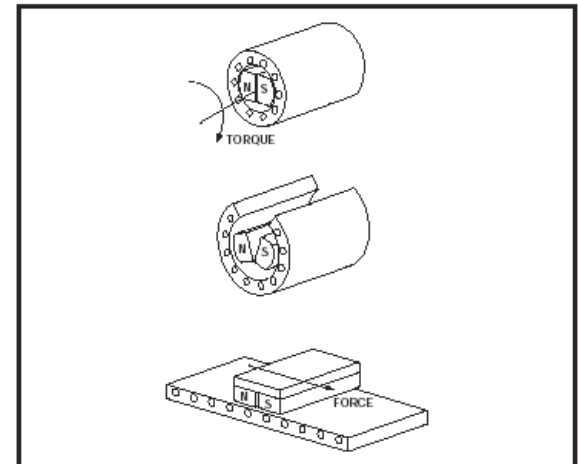


Stepper Motor



Other Concept: Linear Motor

- Linear Motor Technology
- Relative movement
 - Rotor can be stator
 - Stator can be rotor

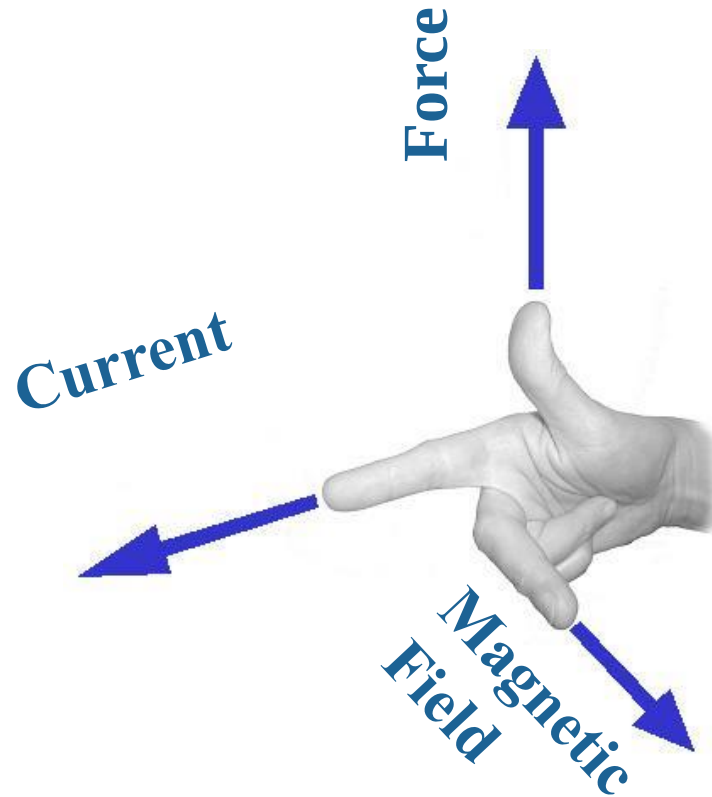


Imaginary process of unrolling a rotary motor.

Traditional DC Machine/Motor

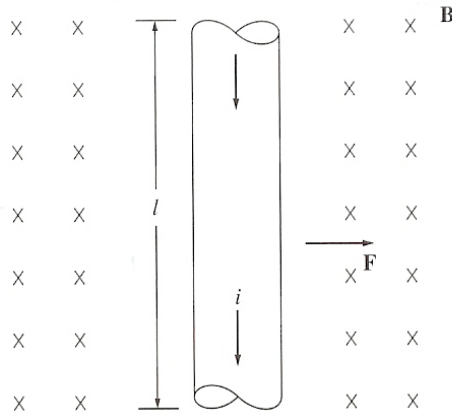
Natural Phenomenon

Electric Currents & Magnetism → Force

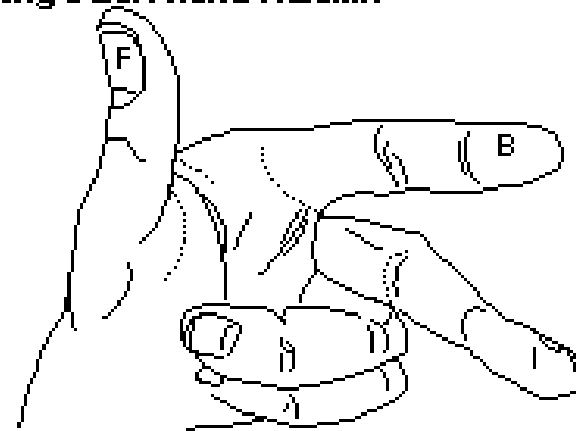


Inducted Force of Current Carrying Conductor in M-field: DC Motor

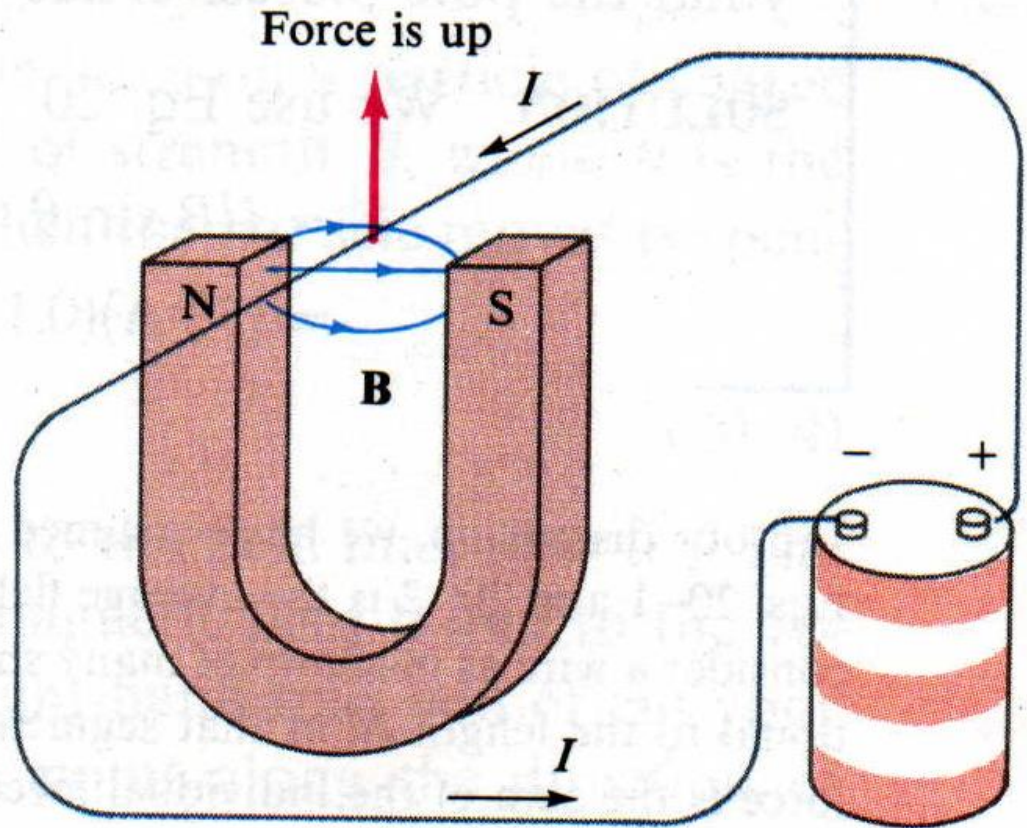
Figure 1-16 | A current-carrying wire in the presence of a magnetic field.



Fleming's Left Hand Rule.....



Force in a Conductor



Force in a Conductor

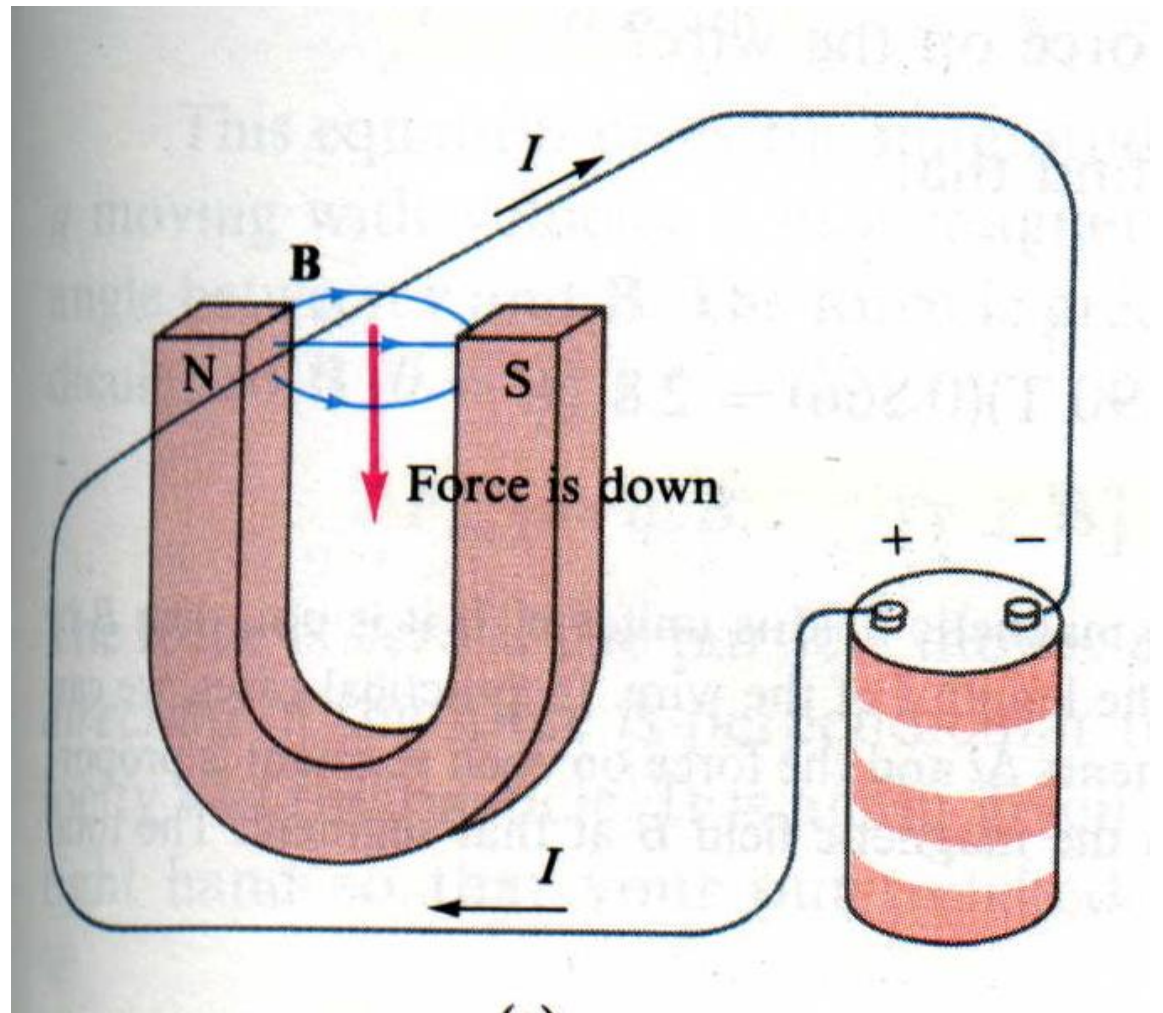
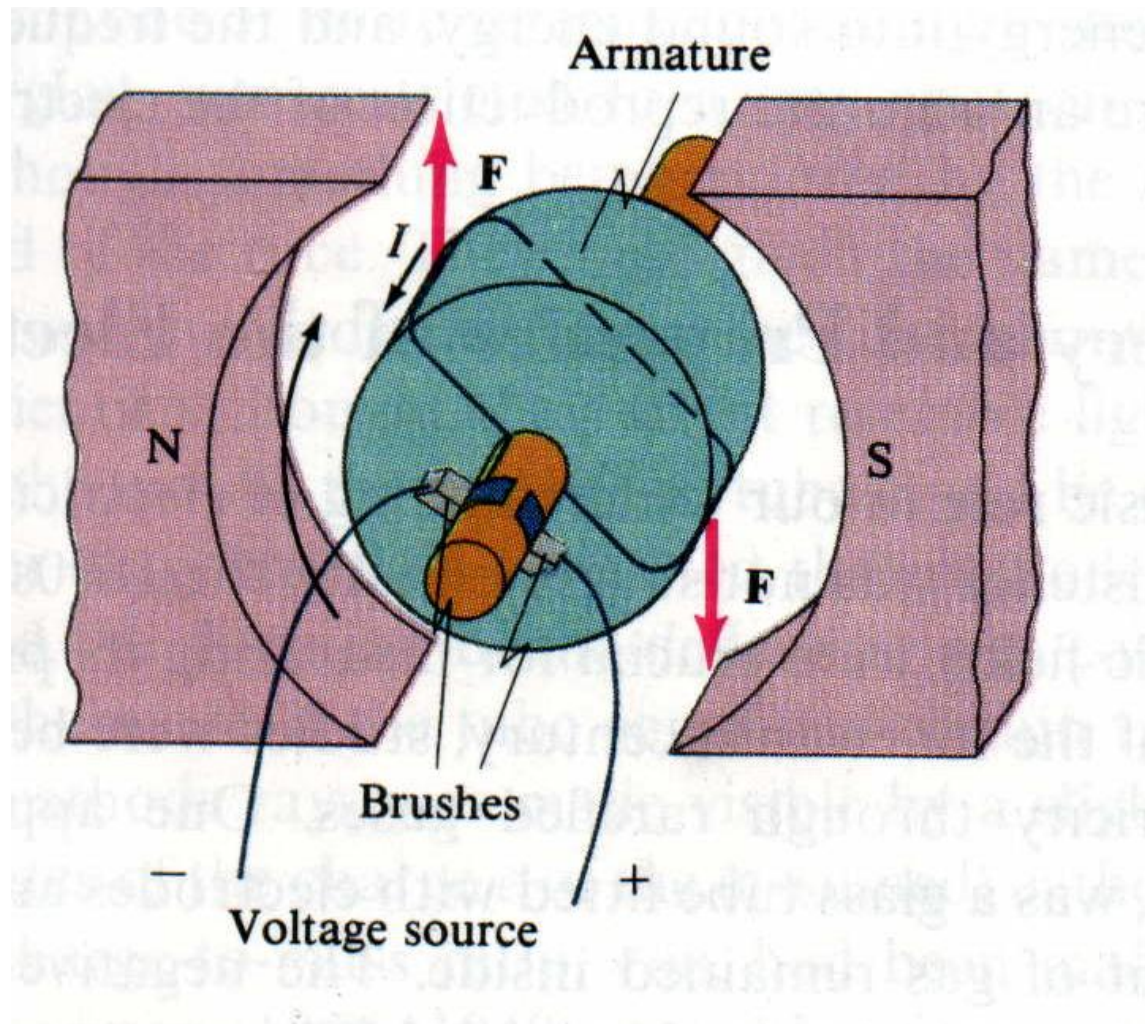
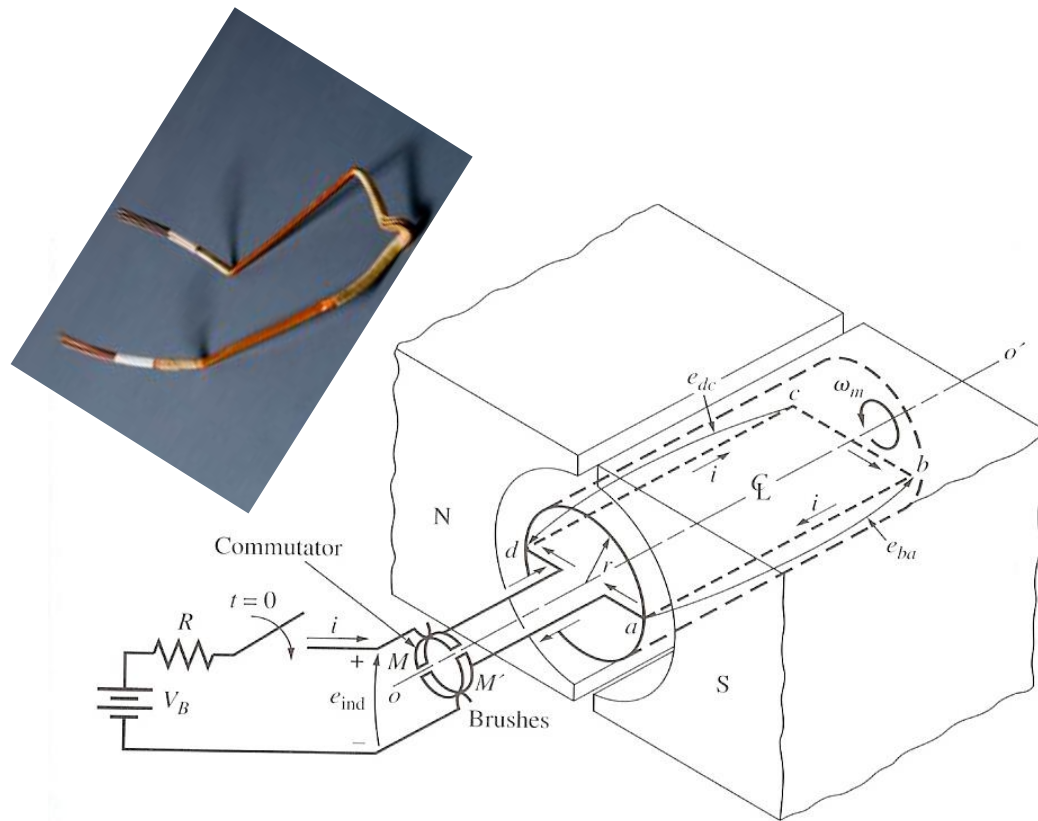


Diagram of a Simple DC Motor

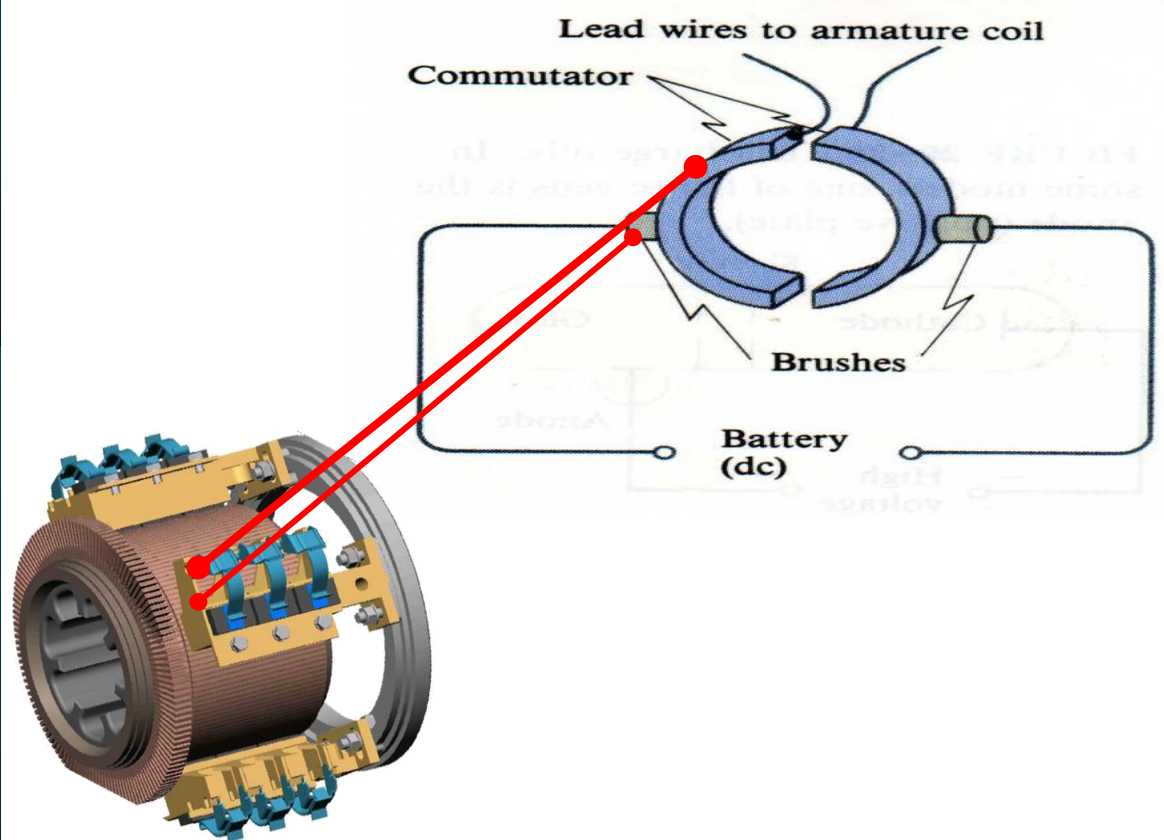


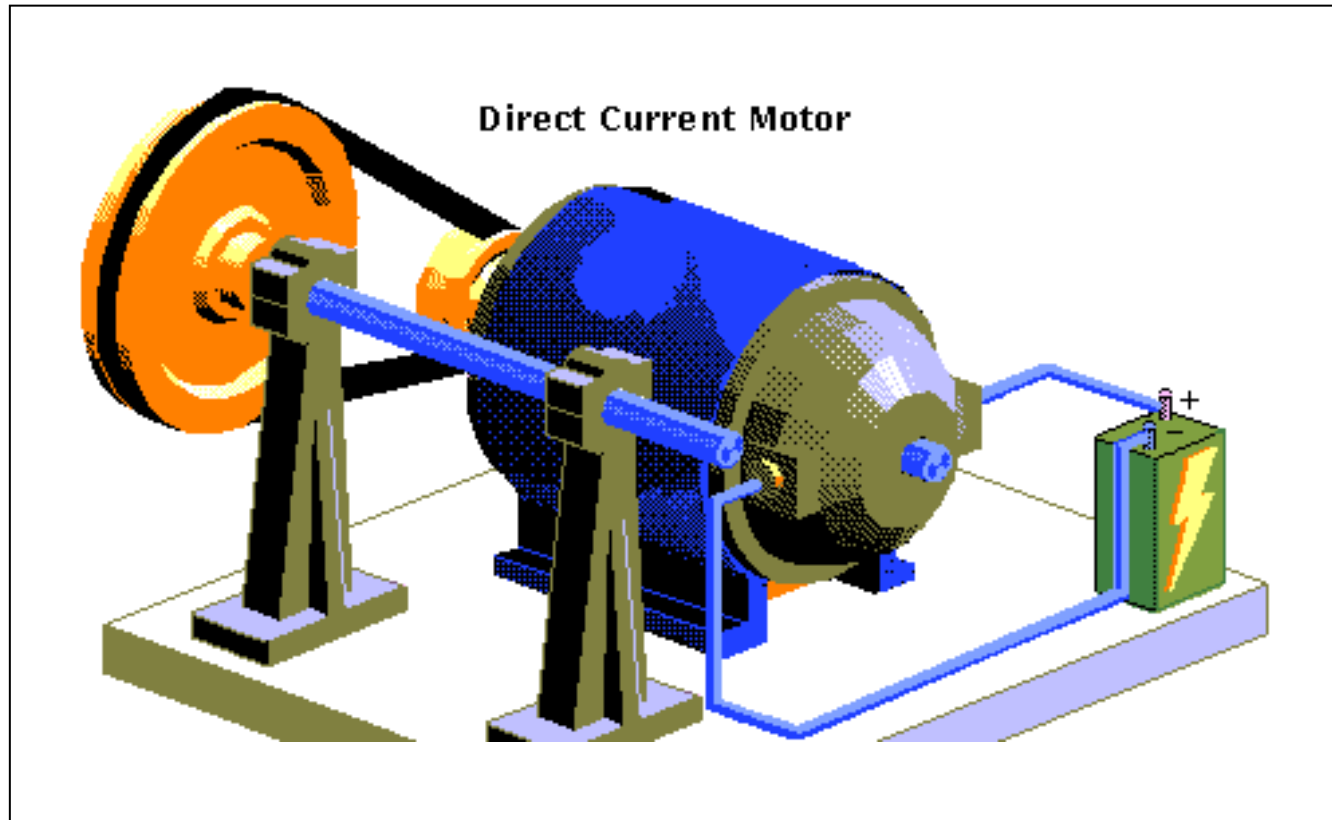
DC Motor Action and Commutation



Commutation in a Simple DC Machine

- What is commutation in a DC machine?
- → *Commutation is the process of switching the loop connections on the rotor of a DC machine just as the voltage in the loop switches polarity, in order to maintain an essentially constant DC output voltage.*





Electric Motor

DC machine and Modeling

- How much torque will be produced in the loop when the switch is closed, and a current is allowed to flow into it?
- To determine the torque, look at the close-up of the loop shown in Figure 8-6b.

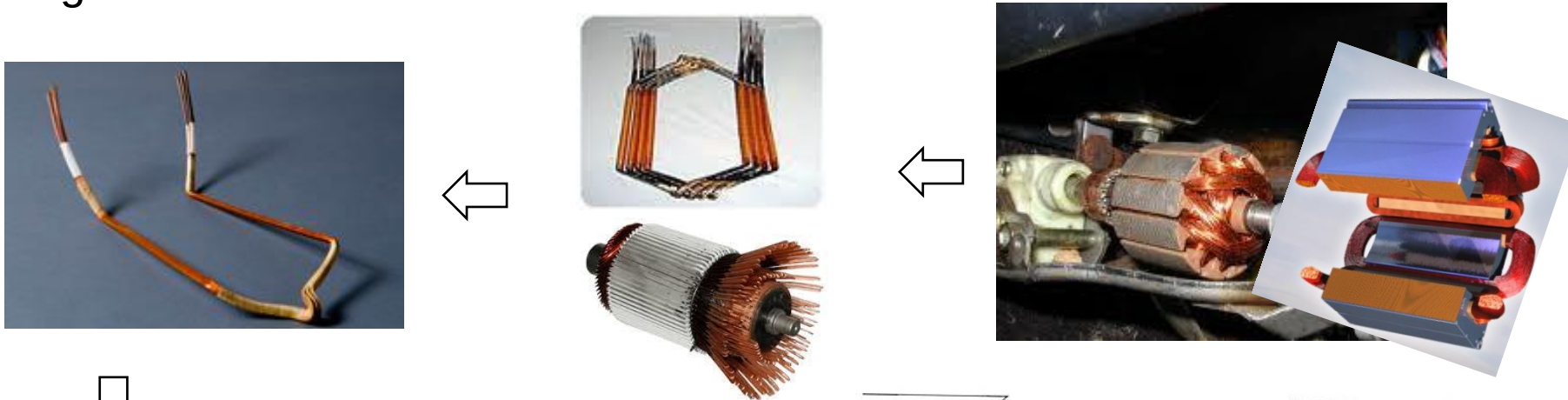
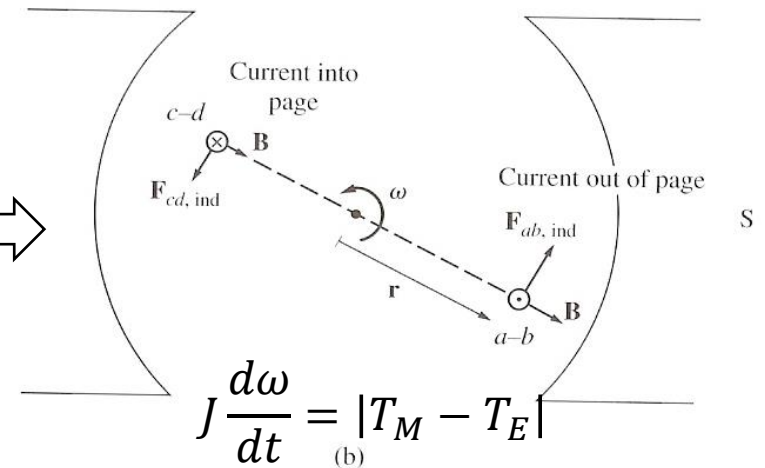
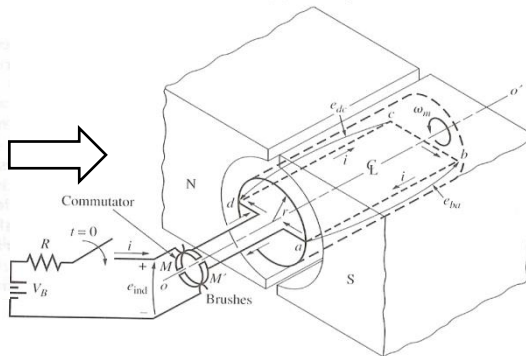
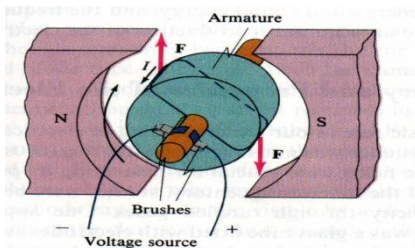
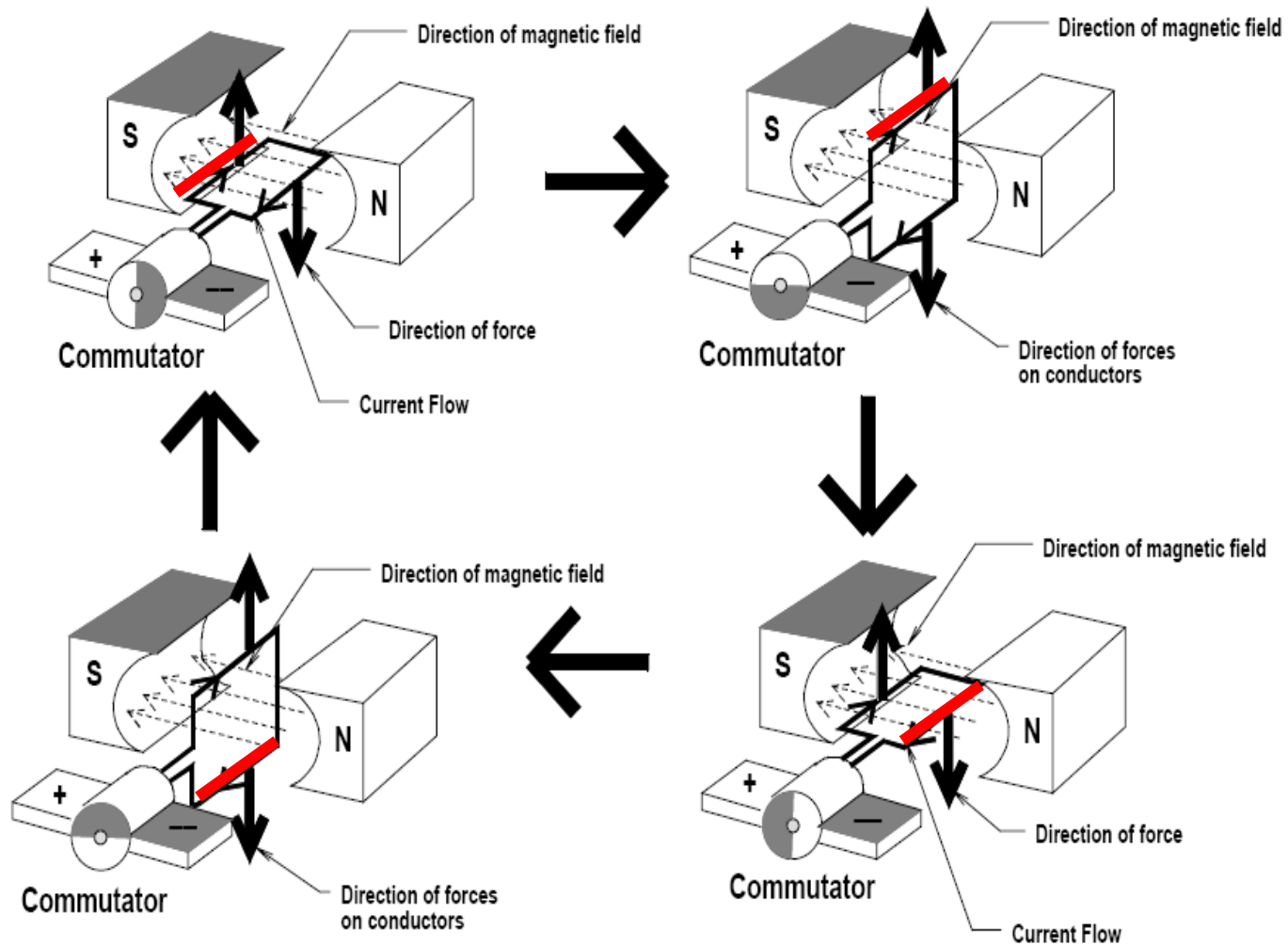


Figure 8-6 | Derivation of an equation for the induced torque in the loop. Note that the iron core is not shown in part (b) for clarity.





Total Torque

- The force on each segment of the loop is given by equation

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B}) \text{ (Fleming's left-hand rule)}$$

- The torque on the segment is given by

$$\tau = rF \sin \theta$$

- Segment ab :

- $F_{ab} = ilB \rightarrow$ tangent to direction of motion
- $\tau_{ab} = rF \sin \theta = r(ilB) \sin 90^\circ = rilB$ (CCW)

- Segment bc :

- $F_{bc} = 0$, Because $\mathbf{l}$ is parallel to $\mathbf{B}$

- Segment cd :

- $F_{cd} = ilB \rightarrow$ tangent to direction of motion
- $\tau_{cd} = rF \sin \theta = r(ilB) \sin 90^\circ = rilB$ (CCW)

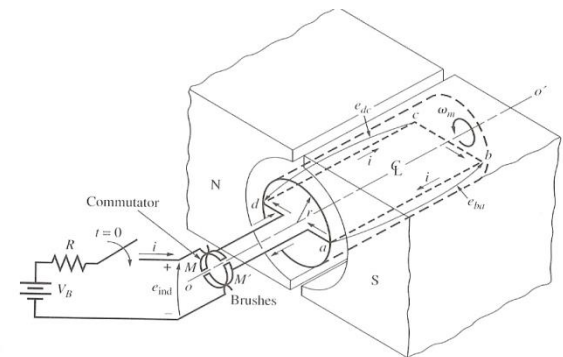
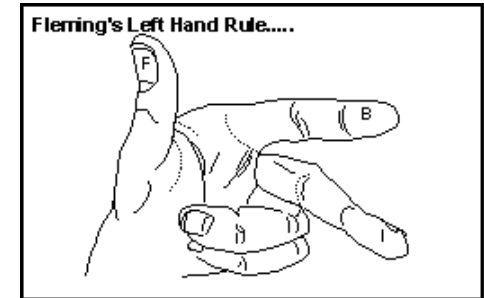
- Segment da :

- $F_{da} = 0$,

- The total induced torque on the loop τ_{ind} thus can be described as,

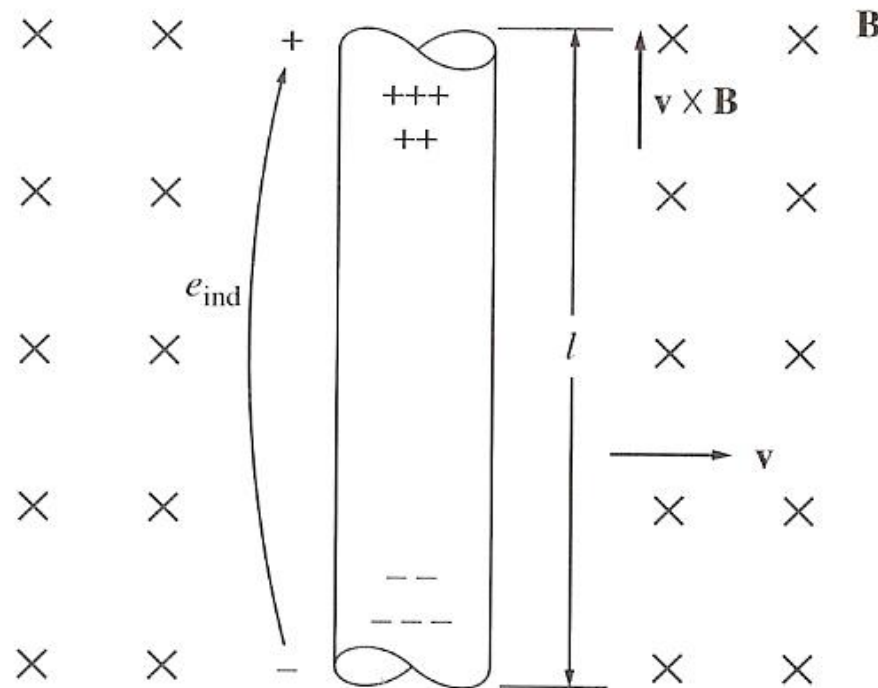
$$\tau_{ind} = \tau_{ab} + \tau_{bc} + \tau_{cd} + \tau_{da} = 2rilB, \text{ and we can easily get,}$$

$$\tau_{ind} = \phi i \text{ or } N\phi i \text{ if we have multiple number of turns}$$

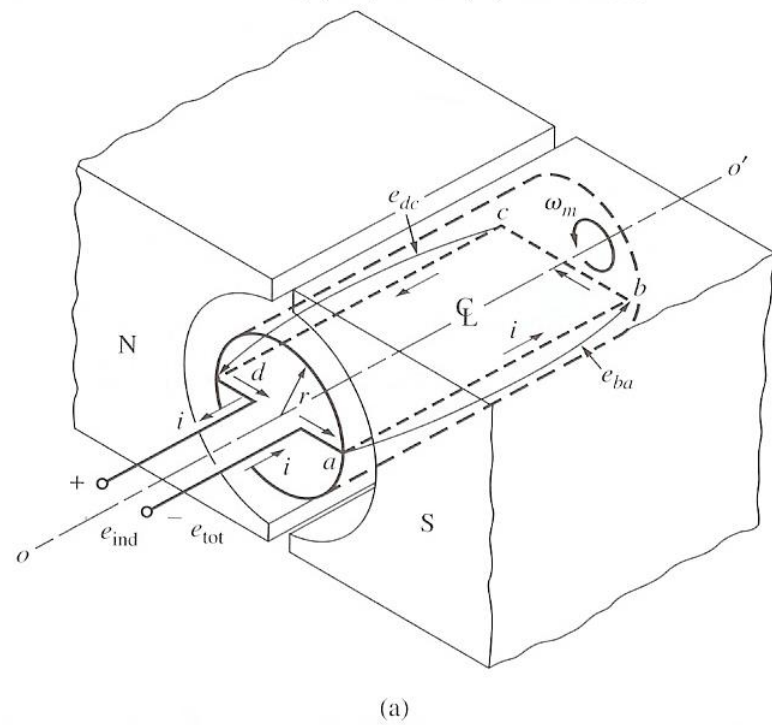


DC Generator

- Faraday's Law, and,
- Induced voltage for a moving conduction in a magnetic field



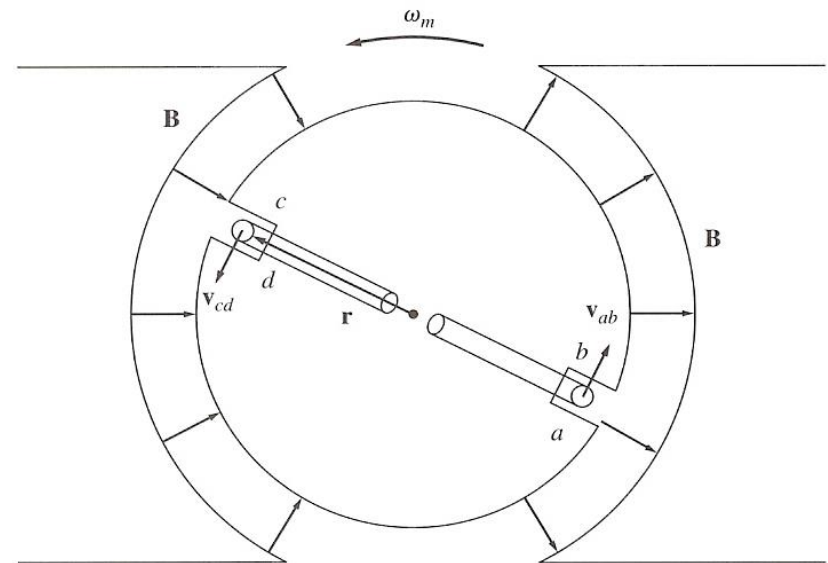
DC Generator Action



Induced Voltages

- The Voltage Induced in a Rotating Loop
 - If the rotor of this machine is rotated, a voltage will be induced in the wire loop → Generator Action

Figure 8–2 | Derivation of an equation for the voltages induced in the loop.



Voltage Magnitude

- The voltage on each segment of the loop is given by equation (Fleming's right-hand rule) $e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$

- Segment ab :

- $e_{ba} = vBl \rightarrow$ positive into page (under the pole faces)
- $e_{ba} = 0$ (beyond the pole edges)

- Segment bc :

- $e_{cb} = 0$, because $(\mathbf{v} \times \mathbf{B}) \perp \mathbf{l}$ (perpendicular)

- Segment cd :

- $e_{dc} = vBl \rightarrow$ positive out of page (under the pole faces)
- $e_{dc} = 0$ (beyond the pole edges)

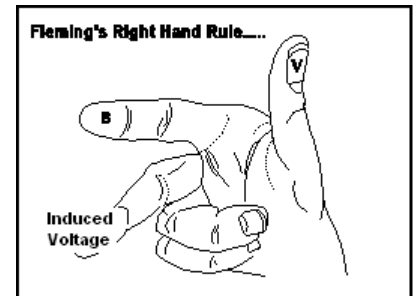
- Segment da :

- $e_{ad} = 0$, because $(\mathbf{v} \times \mathbf{B}) \perp \mathbf{l}$ (perpendicular)

- The total induced voltage on the loop e_{ind} is given by

$$e_{ind} = e_{ba} + e_{cb} + e_{dc} + e_{ad} = 2vBl$$

$$e_{ind} = 2 \frac{d}{dt} \omega Bl = \omega \phi, \quad \text{or } (N)\omega \phi$$



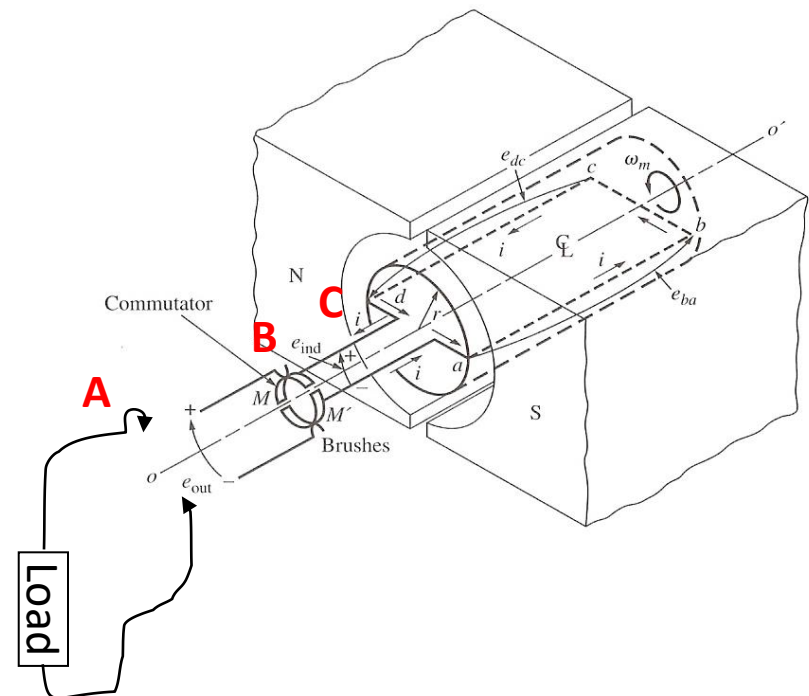
DC Generator Equivalent Circuit

- The magnetic field produced by the stator poles induces a voltage in the rotor (or armature) coils when the generator is rotated.
- This induced voltage is represented by a voltage source.
- The stator coil has resistance, which is connected in series.
- The pole flux is produced by the DC excitation/field current, which is magnetically coupled to the rotor
- The field circuit has resistance and a source
- The voltage drop on the brushes represented by a battery

$$e_{ind} = e_{ba} + e_{cb} + e_{dc} + e_{ad} = vBl + 0 + 0 - (-vBl)$$

$$= 2vBl$$

$$= \omega\phi \quad \text{or} \quad (N\omega\phi)$$

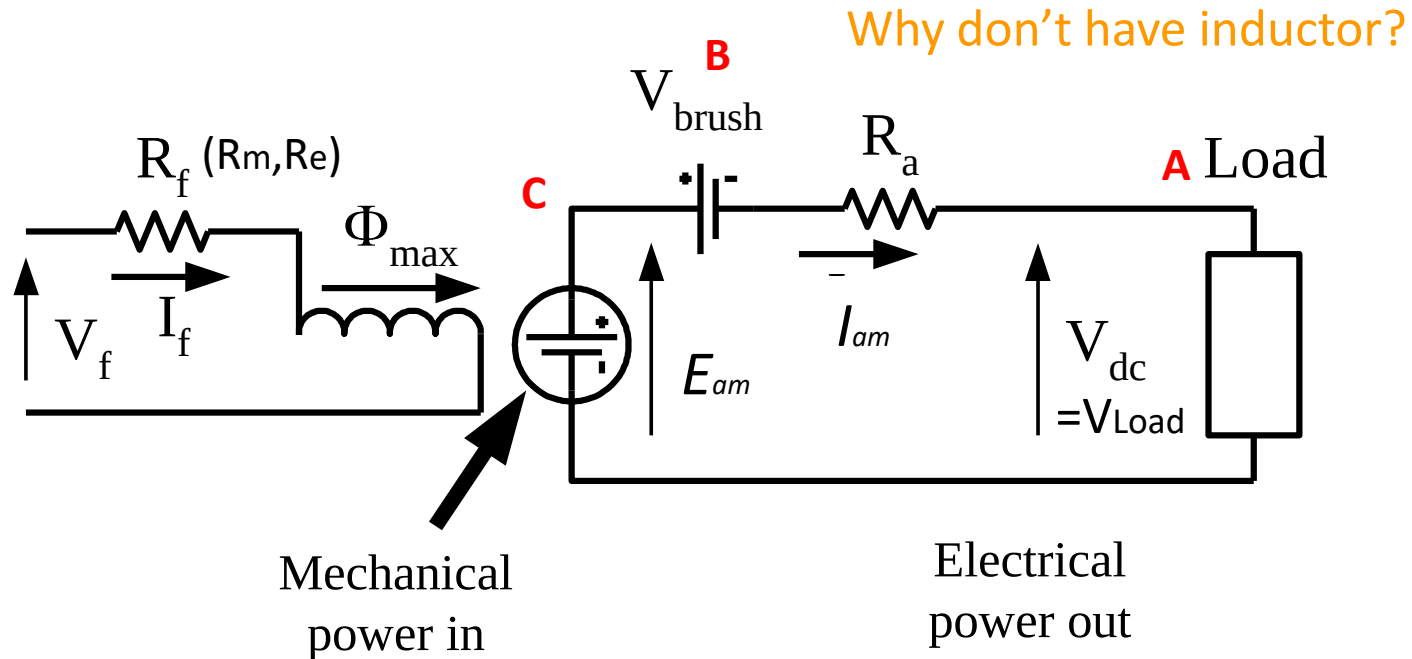


Excitation: DC Generator Equivalent Circuit

- The magnetic field produced by the stator poles induces a voltage in the rotor (or armature) coils when the generator is rotated.
- The dc field current of the poles generates a magnetic flux
- The flux is proportional with the field current if the iron core is not saturated:

$$\Phi_{am} = K_1 I_f$$

DC Generator Equivalent Circuit



$$E_{am} = N_r \Phi_{am} \omega = N_r K_1 I_f \omega = K_m I_f \omega$$

Equivalent circuit of a separately excited DC generator.

DC Generator Equivalent Circuit

- Summary:

$$E_{am} = 2 N_r B \ell_{am} v = 2 N_r B \ell_{am} \left(\omega \frac{D_{am}}{2} \right) = N_r (B \ell_{am} D_{am}) \omega = N_r \Phi_{am} \omega$$

If no saturation

$$\Phi_{am} = K_1 I_f$$

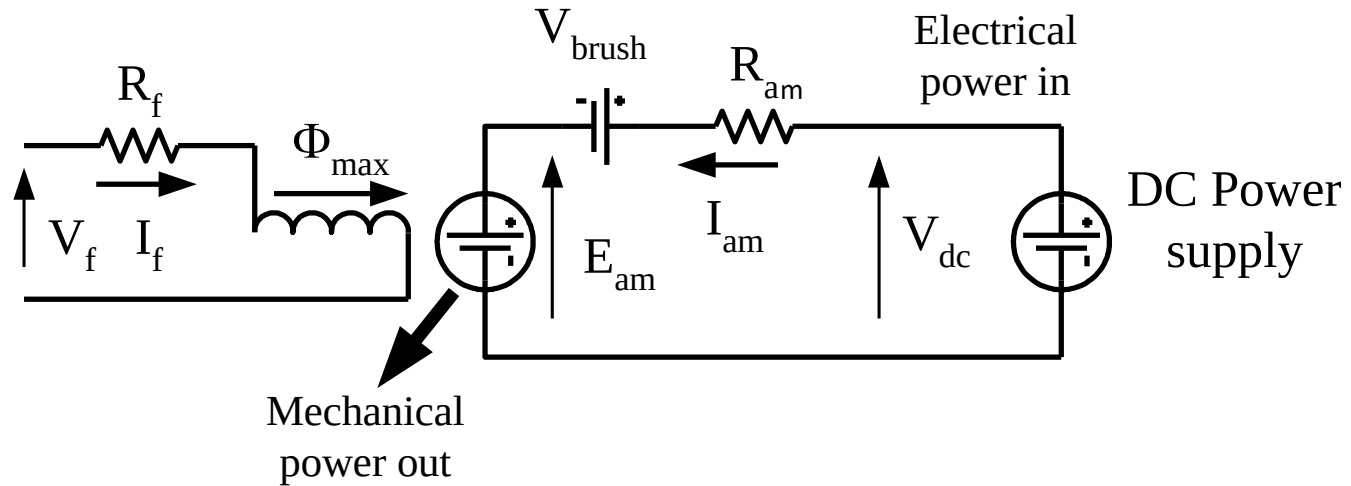
$$E_{am} = N_r \Phi_{am} \omega = N_r K_1 I_f \omega = K_m I_f \omega$$

DC Generator Equivalent Circuit: Generator

- When the generator is loaded, the load current produces a voltage drop on the rotor winding resistance.
- In addition, there is a more or less constant steady state voltage drop on the brushes.
- These two voltage drops reduce the terminal voltage of the generator. The terminal voltage is;

$$E_{am} = V_{Load} + I_{am} R_{am} + V_{brush}$$

DC Motor Equivalent Circuit



- Equivalent circuit of a separately excited dc motor
- Equivalent circuit is similar to the generator only the current directions are different

DC Motor Equivalent Circuit

$$V_{dc} = E_{am} + I_{am} R_{am} + V_{brush}$$

The induced voltage and motor speed vs angular frequency

$$E_{am} = K_m \omega$$

$$K_m = \begin{cases} N\phi, & \text{for permanent magnetic field} \\ Nk_{exe} I_f & \text{for excited/induced magnetic field} \end{cases}$$

$$\omega = 2\pi n_m$$

DC Motor Equivalent Circuit

- The operation equations are:
- The combination of the equations results in

$$K_m \omega = E_{am} = V_{dc} - I_{am} R_{am}$$

The current is calculated from this equation. The steady state output power and torque are: **Note:** $T_{electric} = T_m$

$$P_{out} = E_{am} I_{am} \quad T = \frac{P_{out}}{\omega} = K_m I_{am} I_f$$

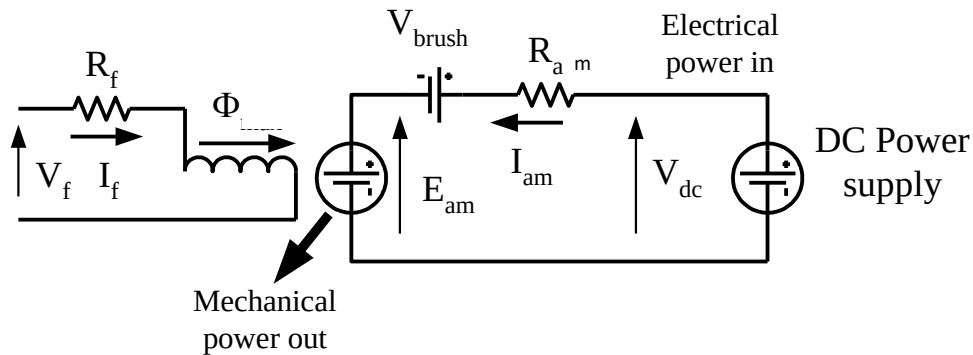
$$P_{in} = V_{dc} I_{am} \quad (\text{also known as } T_{electric})$$

Output Characteristics of DC Motor

- The output variables are shaft torque and speed relationship, also represented by a plot of its output torque versus speed.
- Equivalent circuit of motor
- Induced voltage (angular velocity) and current (torque)
- Equation for DC motor: a equation is just a straight line with a negative slope (see next page for detailed derivation and explanation).

$$\omega = -\frac{R_A}{(KI_f)^2} \tau_{ind} + \frac{V_T}{KI_f}$$

DC Motor Equivalent Circuit



$$N_r \Phi_{am} \omega = E_{am} = V_{dc} - I_{am} R_{am}$$

$$V_{dc} = E_{am} + I_{am} R_{am} + V_{brush} \approx 0$$

$$I_{am} V_{dc} = I_{am} E_{am} + I_{am}^2 R_a + I_{am} V_{brush}$$

$$P_{in} = P_{out} + P_{loss}$$

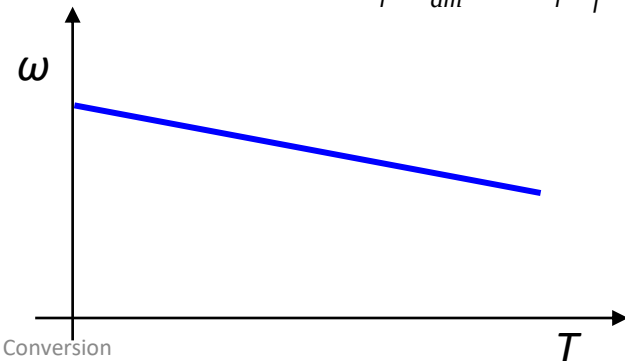
$$P_{out} = E_{am} I_{am} \quad P_{out} = T_m \omega$$

Note: $T_{mechanic} = T$, $T_{electric} = I_{am} \Phi N$

$$T = \frac{P_{out}}{\omega} = N_r \Phi_{am} I_{am} = K_m I_{am} I_f \quad (\Phi_{am} = K_1 I_f, K_m = K_1 N_r) \quad I_{am} = \frac{T}{N_r \Phi_{am}} = \frac{T}{N_r I_f}$$

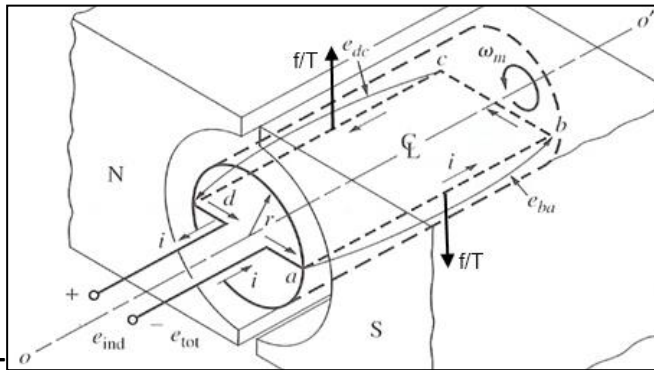
$$V_{dc} = E_{am} + I_{am} R_{am} + V_{brush} = N_r \Phi_{am} \omega + \frac{T R_a}{N_r \Phi_{am}}$$

$$\omega = \frac{V_{dc}}{N_r \Phi_{am}} - \frac{R_{am}}{(N_r \Phi_{am})^2} T$$



Summary

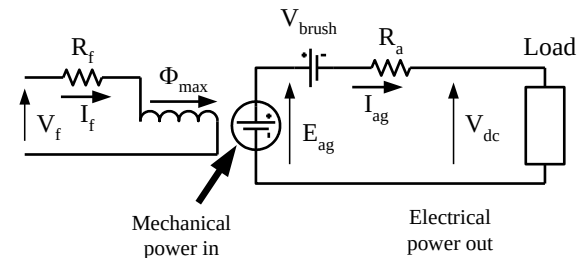
- Structure of DC generator and motor
- Generator and motor Effects
 - Induced voltage (3rd Phenomenon)
 - Induced torque (4th Phenomenon)



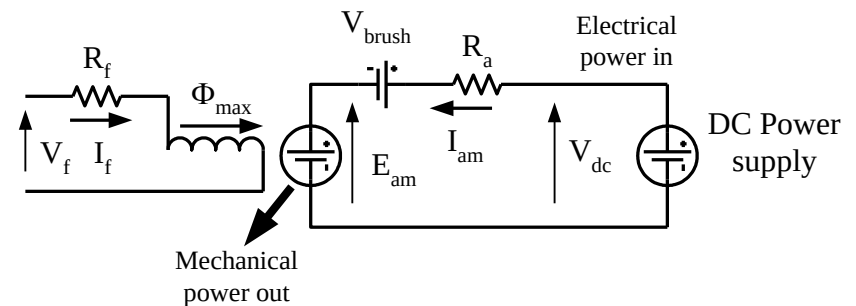
- Flux comes from phenomena 1 & 2
- Power input, output, & efficiency

$$I_{am} V_{dc} = I_{am} E_{am} + I_{am}^2 R_a + I_{am} V_{brush}$$

$$P_{in} = P_{out} + P_{loss}, \quad I_{am} E_{am} = T\omega = P_{out}$$



$$E_{am} = V_{dc} + I_{am} R_{am} + V_{brush} \approx 0$$



$$V_{dc} = E_{am} + I_{am} R_a + V_{brush} \approx 0$$

$$\omega = -\frac{R_a}{(K I_f)^2} \tau_{ind} + \frac{V_T}{K I_f}$$

Analysis of DC Motor

General Equations of DC Motor:

Equivalent Circuit : $V_{dc} = E_{am} + I_{am} R_m + V_{brush}$ (1)

$E_{am} = V_{dc} - I_{am} R_m$ (2), small voltage drop at the brushes

$E_{am} = N_r \Phi_{ag} \omega$ (3), Farady's Law

$T_{am} = N_r \Phi_{ag} I_{am}$ (4), Farady's Law

Power Equations of DC Motor:

Power

$P_{in} = V_{dc} I_{am}$ (4)

$P_{out} = \frac{E_{am} I_{am}}{T_{out} \omega}$ (5)
(electrically)
(mechanically)

$P_{loss} = I_{am}^2 R_m$ (6)

Efficient : $\eta = \frac{P_{out}}{P_{in}} \%$ (7)

Load (Torque/Current) of DC Motor:

Current & torque relation

$\therefore$ Equation (3), (5) & (6) $\therefore I_{am} = \frac{T_{out}}{N_r \Phi_{ag}}$ (8)

Output Characteristics of DC Motor:

$\omega = \frac{V_{dc}}{N_r \Phi_{ag}} - \frac{R_m}{(N_r \Phi_{ag})^2} T_{out}$ (9)

Note : $\omega = 2\pi / 60 = 2\pi(\text{rad / turn}) \times n(\text{turn / min}) \times (1\text{min} / 60)$ (10)

Given:
 $R_a = 0.06\Omega$
 $V_{dc} = 250V$
No-load speed $n_0 = 1200\text{turns/minute}$

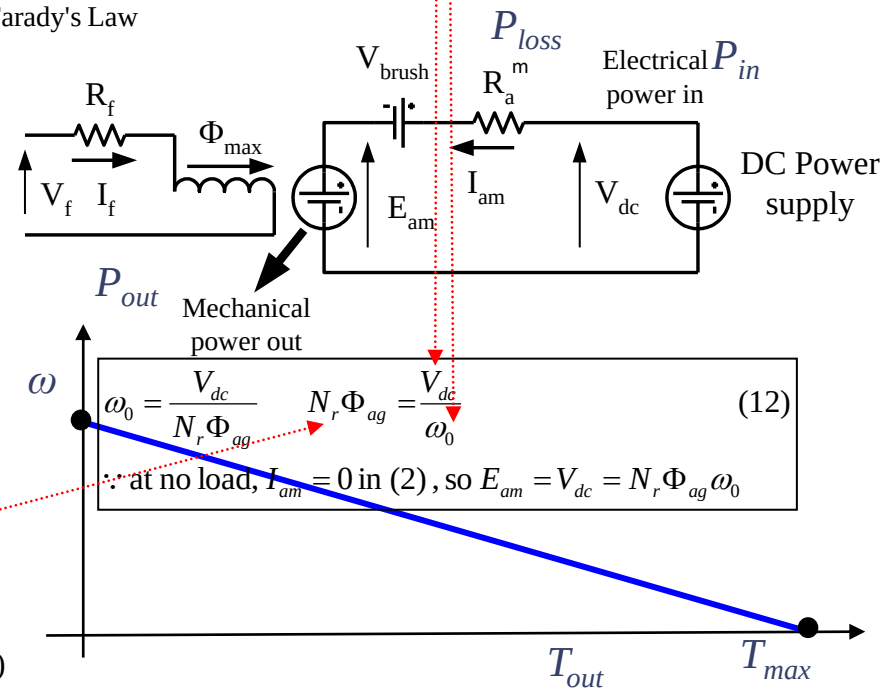
Note: $\omega = 2\pi n / 60$

Find:

T_{load} : (N-m)

Maximum Torque and maximum current

What is the efficiency when $n = 1000\text{turns/minute}$



Hints:

A: T_{max} can be found by letting $\omega = 0$ in Equation (1), and I_{max} can be found using equation (9)

B: Using (10-11) to find T_{out} @ 1000turn/min, then (6) to find P_{out} , (9) to find I_{am} , (4) to find P_{in} , finally using (8) to get efficiency

Course Notes of ECE 3113, Energy Conversion

P_{loss} can also be easily calculated with (7)